# BEBOP: Decision Memo for the Meta Global AI Developer Hackathon

Prepared 3 Nov 2026 | Verdict: NARROW BEBOP | Scores and the 'new' assessment are my estimates, not measured results

**Schedule reality check (verified on the official page).** Project submissions are due **Nov 4, 9:00 PM ET** (Nov 5, 6:30 AM IST); applications closed Oct 25 (or earlier if the 20,000 cap filled). The ten-day window is almost over, so this memo plans **about 24 hours**, not ten days. I found no extension notice. I could not verify whether your account was accepted; you report no Meta credits yet, so a local-model Plan B is in the PRD/TRD/Models documents.

## 1. Thesis and problem

**Thesis.** BEBOP should be a **fail-closed verifier**: it approves a physical action only when each assumption behind it is backed by measured evidence, and otherwise returns UNDECIDED together with the specific missing measurement. **Problem.** Perception-driven systems keep acting on stale or low-quality state without knowing it. The evidence below shows this is real, but it is a family of problems (degraded sensing, stale state, VLM grounding errors, failed actions), not one universal problem.

## 2. Evidence (graded by source type)

| Claim | Type / date | What it shows | Limits |
|---|---|---|---|
| NHTSA EA26002 (upgrade of PE24031) | Official, 18 Mar 2026 | NHTSA frames the issue as failure of FSD's *degradation detection* to detect reduced visibility and warn the driver; about 3.2M vehicles. | Open investigation, no defect finding or recall in sources I read. Crash counts (9, one fatal) come from secondary press. Does not prove an AI-perception root cause. |
| VAP-TAMP (arXiv 2604.26988) | Preprint, 28 Apr 2026 | Says VLMs fail under occlusion and give no calibrated confidence. 72 of 100 real trials hit at least one unexpected situation; success 88% vs 76% (OK-Robot) and 69% (COWP). | Authors' own results, one lab, Gemini as VLM. |
| AsgardBench (arXiv 2603.15888) | Preprint, 16 Mar 2026 | Text-only drops to 36.1% accuracy; reports weak visual grounding and state tracking in VLM planners. | Simulator, high-level actions only. |
| Dream2Fix (arXiv 2603.13528) | Preprint, 13 Mar 2026 | Fine-tuned VLM raises correction accuracy 19.7% to 81.3%; 46% zero-shot real-robot recovery. | Needs fine-tuning data; out of reach in 24 h. |
| Reddit thread on identity under occlusion | Anecdote, Mar 2026 | A practitioner says ReID mostly solves occlusion except for identical objects. | 3 upvotes; a lead, not evidence. |

**Coverage gap.** I did not get meaningful X/Twitter, GitHub-issue or Hacker News coverage, and Reddit gave one weak lead. I could not locate primary sources matching CLOSER, Vision-Guided Recovery or constrained natural-language action planning, so they are not cited. The 'budgeted replanning' paper (arXiv 2608.01428) was seen by title only. Do not cite Tesla as proof that perception caused crashes.

## 3. Closest prior work and the remaining gap

| System | Detects change | Diagnoses | Requests new evidence | Replans | Overlap with original BEBOP |
|---|---|---|---|---|---|
| VAP-TAMP (2026) | Yes (pre/post-condition VQA) | Perception error vs world change vs execution failure | Yes, VLM-chosen viewpoint (1.61 vs 4.35 views) | Yes (PDDL) | **Very high**: most of the loop |
| Ahmad et al. (2025) | Yes | VLM finds missing condition/skill | No | Yes (behavior trees) | High |
| RoVLaP (AAMAS 2026) | Via belief states | No | Active sensing listed as future work | Plans across plausible states | Medium |
| AP-VLM (2024), RePLan (2024) | Partial | Partial | AP-VLM: VLM picks viewpoint | RePLan: yes | Medium |
| AsgardBench (2026) | Benchmark | Benchmark | n/a | n/a | Evaluation reference only |

**What is genuinely new?** Not the pipeline: change detection, diagnosis, VLM-guided extra observation and replanning already exist in VAP-TAMP. **Red-team result:** a team that built VAP-TAMP could argue that its paraphrase-vote check, view-sufficiency check and re-verification of preconditions before every action already cover BEBOP's loop, and that re-verifying before acting makes explicit dependency tracing largely redundant. I accept that for the ledger, so it is now an explanation aid, not a novelty claim. What I found in the paper (code not audited) is that its Algorithm 2 still returns a binary value when the viewpoint budget runs out, so it has no abstain output, and its checks are VLM yes/no rather than metric. The defensible gap is therefore narrow: **a verifier that abstains with a quantified margin, tested head-to-head on scenes where the evidence is genuinely insufficient**. If a VAP-style baseline matches it on false approvals, the contribution shrinks to engineering and should be reported that way.

**Verdict on the problem:** real, important, and worth solving in narrow form. **Original BEBOP is not differentiated enough to build as specified.** First users: robotics/CV engineers who need an audit trail for 'why did the system say go?'. Today they use ad-hoc thresholds and logs.

## 4. Recommended solution: BEBOP-Gate (tabletop clearance verifier)

**Workflow.** (1) SAM 3.1 segments the moving object, obstacles and passage edges from a short clip. (2) Deterministic code measures the gap and its uncertainty (spread across 3 frames x 5 prompt variants, plus a printed scale marker). (3) Each assumption (for example *gap at least object width + margin*) is written to a versioned **ledger** with the mask evidence that supports it. (4) New frames re-score every assumption; if an input changed, the assumption and every decision that depends on it are marked invalid. (5) Verdict: PASS only if margin exceeds 2 sigma; FAIL if the violation exceeds 2 sigma; otherwise UNDECIDED plus the missing measurement (which dimension, which tolerance, which unoccluded view). (6) If FAIL, check an alternate route with the same verifier.

| Component | Handled by | Why |
|---|---|---|
| Object/passage masks, tracking across frames | SAM 3.1 (docs) | Text prompt in, box + mask out; video tracking documented. Output is special-token text that needs parsing (not yet tested by me). |
| Semantic labelling of ambiguous scenes, plain-language explanation | Muse Spark 1.3 (Responses API) | Only place free-text reasoning helps. Never decides PASS/FAIL. |
| Gap geometry, uncertainty, tri-state verdict, ledger, invalidation | Deterministic Python | Must be auditable and reproducible; an LLM here would reintroduce false approvals. |

**Illustrative trace (not measured).** t0: gap 142 +/- 6 mm vs object 120 mm + 5 mm margin -> PASS; ledger A1 cites masks m1-m3. t1: a block moves, blocker mask changes -> A1 invalid, route R1 flagged. t2: hand occludes edge, gap 128 +/- 14 mm -> UNDECIDED; missing measurement: edge width to +/- 4 mm from an unoccluded top-down view. t3: new frame, gap 112 +/- 3 mm -> FAIL; route R2 verified PASS; explanation cites the evidence.

## 5. Evaluation (build this; report only what you measure)

**Data:** about 40 phone-captured scenes: 3 gap classes (clear pass, marginal, clear fail) x clean, hand-occluded, blur, low light, lookalike distractor, object moved mid-clip. At least 10 are **unanswerable by design** (the deciding edge is hidden or the gap is within measurement error), so the correct output is UNDECIDED. **Ground truth:** ruler/caliper gap. **Baseline 1:** Muse Spark yes/no on the image. **Baseline 2:** SAM mask + geometry, single shot, binary. **Baseline 3 (new):** a VAP-style verifier I reimplement from the paper's description on Muse Spark (5 paraphrased questions, majority vote, sufficiency check, forced binary answer at budget exhaustion); label it a reimplementation, not their code. **Metrics, in priority order:** false-approval rate on all scenes and on the unanswerable subset; risk-coverage curve (errors vs fraction answered); stale-plan retention after a change; missed changes; UNDECIDED cases resolved by one extra measurement. **Target, not result:** fewer false approvals than all three baselines at at least 70% coverage.

## 6. Alternatives compared (my 1-10 estimates; higher is better, 'Low-dup' = low duplication risk)

| Direction | Sev | Dem | Orig | Depth | Feas | Meta | Demo | Eval | Low-dup | Cred | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|
| A. Decision-history + recovery | 6 | 5 | 3 | 6 | 6 | 6 | 7 | 6 | 3 | 6 | 54 |
| B. Active perception, next observation | 6 | 5 | 3 | 7 | 4 | 6 | 6 | 5 | 3 | 4 | 49 |
| C. Plan verifier + smallest missing measurement | 7 | 6 | 5 | 6 | 8 | 7 | 8 | 9 | 5 | 8 | **69** |
| D. Perception-degradation gate (from NHTSA evidence) | 8 | 7 | 4 | 5 | 8 | 6 | 7 | 9 | 4 | 8 | 66 |

Reasoning: A and B are the most duplicated by VAP-TAMP and B needs camera motion or hardware. C is the most objectively testable. D has the strongest official evidence but is mostly uncertainty estimation. **Recommendation merges C and D**: D's instability signal produces UNDECIDED; C produces the verdict.

## 7. 24-hour plan, risks, hackathon fit

| Hours | Deliverable | Fallback if blocked |
|---|---|---|
| 0-2 | Confirm acceptance and key; SAM 3.1 smoke test; parse masks | No credits: local Grounding DINO + SAM 2.1 (Apache-2.0) or OpenCV color segmentation; label substitutes honestly |
| 2-8 | Capture 40 scenes with ground truth; gap + uncertainty code | Cut to 24 scenes |
| 8-14 | Tri-state verifier, ledger, invalidation | Drop alternate-route check |
| 14-18 | Run both baselines + BEBOP; table of measured results | Report fewer metrics, never invented ones |
| 18-22 | Simple Streamlit/HTML view: overlay, ledger, verdict | Static annotated images |
| 22-24 | README, 2-minute demo video, submit with 3 h buffer before 9 PM ET | Submit the working subset |

**Reuse (licenses from memory, verify before use):** OpenCV (Apache-2.0), Shapely (BSD-3), supervision (MIT, optional overlays). **Too ambitious for 24 h:** belief-space planning, VLM-driven viewpoint robots, Dream2Fix-style fine-tuning, information-gain optimization. **Do not build:** 3D reconstruction, a simulator, multi-agent orchestration, robot control. **Cost** (pricing): SAM $2.50 per 1,000 images, $0.20 per 1,000 video frames (50-frame minimum per request); Contributor mode trains Meta models on your content.

**Fit (Indie track):** works-and-measured is the strong point; Push the Frontier fits via robotics/physical-system verification; Build for Your World is weak unless you have a real domain, so do not force it. **Crowded field:** public repos BONISA, SceneProof and LandProof already pair Meta models with deterministic checks or abstention in other domains (README openings only), so lead with the measured false-approval table. **Biggest risks:** API access or SAM output parsing; too few scenes; a judge seeing a VLM wrapper (mitigated by the deterministic core and measured false-approval table); Baseline 3 matching BEBOP, which must be reported honestly.

**FINAL RECOMMENDATION: NARROW BEBOP** to the tri-state clearance verifier above. If you are not an accepted participant, you cannot enter this event; build it anyway as an open-source portfolio piece. **Single next action (next hour):** run one SAM 3.1 call on a tabletop photo, parse the mask, and print a gap width in millimetres.